Lecture 1

HHU summer term 2026

April 15, 2026

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Predictive Inference with Linear Regression in Moderately High Dimensions

Foundation of Linear Regression

- A linear regression model is given by

$$Y = \beta' X + \varepsilon,$$

  where ε is a so called noise term with $\mathbb{E}[\varepsilon] = 0$.

- We consider a scalar random variable Y, an outcome of interest, and a p-dimensional vector of covariates $X = (X_1, \ldots, X_p)'$. We assume that a constant of 1 is included as the first component in X; that is, $X_1 = 1$.

- Our first goal is to construct the best linear prediction rule for Y using X. That is, the predicted value of Y given X will be of the linear form:

$$\sum_{j=1}^{p} \beta_j X_j = \beta' X, \text{ for } \beta = (\beta_1, \ldots, \beta_p)'.$$

Best Linear Predictor

- The β's are called the regression parameters or coefficients.
- We define β as any solution to the **Best Linear Prediction** (BLP) Problem,

$$\min_{b \in \mathbb{R}^p} \mathbb{E}\left[(Y - b'X)^2\right],$$

where we minimize the Expected or Mean Squared Error (MSE) for predicting Y using the linear rule

$$b'X = \sum_{j=1}^{p} b_j X_j, \quad b = (b_1, \ldots, b_p)'.$$

- The solution to this optimization problem, $\beta' X$, is called the **Best Linear Predictor** of Y using X. This jargon refers to the fact that $\beta' X$ is the best, according to Mean Squared Error, linear prediction rule for Y among all possible linear prediction rules.

Normal Equations

- We can compute an optimal β by solving the first order conditions for the BLP problem:

$$\mathbb{E}[(Y - \beta' X)X] = 0.$$

  These equations are also referred to as the Normal Equations and are obtained by setting the derivative of the objective function $b \mapsto \mathbb{E}(Y - b'X)^2$ with respect to b equal to zero. Thus, any optimal $b = \beta$ satisfies the Normal Equations (*Moment Condition*).

- Noting that $\varepsilon = Y - \beta' X$, we can write the Normal Equations as

$$\mathbb{E}[\varepsilon X] = 0 \text{ or } \varepsilon \perp X.$$

- Therefore, we obtain the simple decomposition of Y:

$$Y = \beta' X + \varepsilon, \quad \mathbb{E}[\varepsilon X] = 0,$$

  where $\beta' X$ is the part of Y that can be linearly predicted or explained with X, and ε is the remaining unexplained or residual part.

- What about the assumption $E[\varepsilon | X] = 0$?

The Best Linear Prediction Problem in Finite Samples

- In practice, the researcher does not have access to the entire population, but observes only a sample

$$(Y_i, X_i)_{i=1}^n = ((Y_1, X_1), \ldots, (Y_n, X_n)).$$

- We assume that this sample is a random sample from a distribution F, which is the distribution of (Y, X). Formally, this condition means that the observations were obtained as realizations of independently and identically distributed (iid) copies of the random variable (Y, X).
- We construct the best in-sample linear prediction rule for Y using X analogously to the population case by replacing theoretical expected values, $\mathbb{E}$, with empirical averages $\mathbb{E}_n$ (*sample mean*), where

$$\mathbb{E}_n[f(Y_i, X_i)] := \frac{1}{n} \sum_{i=1}^n f(Y_i, X_i).$$

- Specifically, given X, our predicted value of Y will be

$$\sum_{j=1}^{p} \hat{\beta}_j X_j = \hat{\beta}' X, \text{ for } \hat{\beta} = (\hat{\beta}_1, \ldots, \hat{\beta}_p)',$$

  where $\widehat{\beta}$ is any solution to the **Best Linear Prediction Problem in the Sample**:

$$\min_{b \in \mathbb{R}^p} \mathbb{E}_n[(Y_i - b' X_i)^2].$$

  That is, $\hat{\beta}$ minimizes the Sample Mean Squared Error for predicting Y using the linear rule $b' X$.

- The $\hat{\beta}$ is called OLS (*ordinary least squares*) estimator.

Properties of Sample Linear Regression

- A key question is whether $\hat{\beta}' X$ approximates (that is, estimates) $\beta' X$ well.
- The fundamental statistical issue is that we are trying to estimate p parameters, $\beta_1, \ldots, \beta_p$. Intuitively, to estimate each parameter well, we need many observations per parameter. This intuition suggests that p/n should be small, in order for estimation error to be small. The following result captures this intuition more formally.

Theorem 1 (Approximation of BLP by OLS)
Under regularity conditions,

$$\sqrt{\mathbb{E}[(\beta' X - \hat{\beta}' X)^2]} \leq \mathrm{const}_F \cdot \sqrt{\mathbb{E}[\varepsilon^2]} \sqrt{\frac{p}{n}},$$

where the inequality holds with probability approaching 1 as $n \to \infty$, and const_F is a constant that depends on the distribution F of (Y, X).

Goodness of Fit

- We can use the sample/population MSE (Mean Squared Error),

$$\mathrm{MSE}_{sample} = \mathbb{E}_n[\hat{\varepsilon}_i^2], \quad \mathrm{MSE}_{pop} = \mathbb{E}[\varepsilon^2]$$

  or the sample/population R^2,

$$R^2_{sample} := \frac{\mathbb{E}_n[(\hat{\beta}' X_i)^2]}{\mathbb{E}_n[Y_i^2]} = 1 - \frac{\mathbb{E}_n[\hat{\varepsilon}_i^2]}{\mathbb{E}_n[Y_i^2]} \in [0,1], \quad R^2_{pop} := \frac{\mathbb{E}[(\beta' X)^2]}{\mathbb{E}[Y^2]}.$$

  to measure the quality of our (linear) model. It is worth noting that R^2_{pop} is the proportion of variation of Y explained by the BLP

- By the law of large numbers and Theorem 1, when p/n is small, we have the following approximations:

$$\mathbb{E}_n[Y_i^2] \approx \mathbb{E}[Y^2], \quad \mathbb{E}_n[(\hat{\beta}' X_i)^2] \approx \mathbb{E}[(\beta' X)^2], \quad \mathbb{E}_n[\hat{\varepsilon}_i^2] \approx \mathbb{E}[\varepsilon^2],$$

- Thus, when p/n is small and n is large, the sample fit measures are good approximations to population fit measures:

$$\mathrm{MSE}_{sample} \approx \mathrm{MSE}_{pop} \text{ and } R^2_{sample} \approx R^2_{pop}.$$

Overfitting: What happens when p/n is not small?

- When p/n is not small, the picture about predictive performance of the in-sample BLP becomes inaccurate and possibly misleading.
- Consider an extreme example where $p = n$ and all variables in X are linearly independent. In this case, we have

$$\mathrm{MSE}_{sample} = 0 \text{ and } R^2_{sample} = 1$$

  no matter what MSE_{pop} and R^2_{pop} are. **Why?**
- Therefore, here we have an extreme example of **overfitting**, where the in-sample predictive performance overstates the out-of-sample predictive performance of the linear model.

- Example:

Suppose $X \sim N(0, I_p)$ and $Y \sim N(0,1)$ are statistically independent.

```r
set.seed(123)
n <- 100
p <- n
ratio <- p/n
cat("The ratio (p/n) is", ratio, "\n")
```

```
The ratio (p/n) is 1
```

```r
X <- matrix(rnorm(n*p), n, p)
Y <- rnorm(n)
model <- lm(Y~X)
cat("R2 is",summary(model)$r.squared, "\n")
```

```
R2 is 1
```

It holds $R^2_{sample} = 1$ but $R^2_{pop} = 0$, since the best linear predictor of Y is $\beta' X = 0$.

- In the tutorial, we will consider an extensive simulation study to understand overfitting, see here.

Measuring Predictive Ability by Sample Splitting

- How should we measure the predictive ability of the linear model (or other nonlinear models that we discuss) more reliably, even in cases when p/n is not small?

A general way to measure predictive performance is to perform **data splitting**. The idea can be summarized in two parts:

1. Use a random part of a dataset, called the training sample, for estimating/training the prediction rule.
2. Use the other part, called the testing sample, to evaluate the quality of the prediction rule, recording out-of-sample mean squared error and R^2.

- Generally, a predictive model is trained on a sample and the real test of its predictive ability happens when "new, unseen" observations arrive. With new observations in hand, we learn how far off our predictions where then compared to the realized values. By partitioning the data set into two parts, we preserve an "unseen" set of observations on which to test our model, mimicking this process of ex-post performance assessment.

Inference about Predictive Effects or Association

- Here we examine inference on so-called predictive effects, which describe how our (population best linear) predictions change if a value of a target regressor changes by a unit while the other regressors remain unchanged.
- Specifically, we partition the vector of regressors X into two components:

$$X = (D, W')',$$

  where D represents the "target" regressor of interest, and W represents the other regressors, sometimes called the controls.
- We can therefore write

$$Y = \beta_1 D + \beta_2' W + \varepsilon.$$

"How does the predicted value of Y change if D increases by a unit while W remains unchanged?"

- The answer is the predicted value of Y changes by β_1.
- Note that this question is purely about the properties of the predictive rule, and generally has nothing to do with causality.
- **Example** (Gender Pay Gap): In the analysis of wages, which we will discuss later in more detail, the question can be formulated as: "What is the difference in predicted wages between men and women with the same job-relevant characteristics?"
- Here, D would represent the female indicator and W represent experience, educational, occupational, and geographic characteristics, the answer to the question is the population regression coefficient

$$\beta_1$$

corresponding to D.

Understanding β_1 via "Partialling-Out"

- "Partialling-out" is an important tool that provides conceptual understanding of the regression coefficient β_1.
- In the population, we define the partialling-out operation as a procedure that takes a random variable V and creates a "residual" $\widetilde{V}$ by subtracting the part of V that is linearly predicted by W:

$$\widetilde{V} = V - \gamma_{VW}' W, \quad \gamma_{VW} \in \arg\min_{\gamma} \mathbb{E}[(V - \gamma' W)^2].$$

- When V is a vector, we apply the operation to each component. It can be shown that the partialling-out operation is linear in the sense that

$$Y = V + U \implies \widetilde{Y} = \widetilde{V} + \widetilde{U}.$$

- We apply the partialling-out operation to both sides of our regression equation

$$Y = \beta_1 D + \beta_2' W + \varepsilon$$

to get

$$\widetilde{Y} = \beta_1 \widetilde{D} + \beta_2' \widetilde{W} + \tilde{\varepsilon},$$

which simplifies to the decomposition:

$$\widetilde{Y} = \beta_1 \widetilde{D} + \varepsilon, \quad \mathbb{E}[\varepsilon \widetilde{D}] = 0.$$

- **Why** does $\widetilde{W} = 0$ and $\tilde{\varepsilon} = \varepsilon$ hold?
- Therefore, β_1 can be interpreted as a (univariate) linear regression coefficient in the linear regression of *residualized* Y on *residualized* D, where the residuals are defined by partialling-out the linear effect of W from Y and D.

Theorem 2 (Frisch-Waugh-Lovell, Frisch and Waugh (1933), Lovell (1963), Lovell (2008))
The population linear regression coefficient β_1 can be recovered from the population linear regression of $\widetilde{Y}$ on $\widetilde{D}$:

$$\beta_1 = \arg\min_{b_1} \mathbb{E}[(\widetilde{Y} - b_1 \widetilde{D})^2] = (\mathbb{E}[\widetilde{D}^2])^{-1} \mathbb{E}[\widetilde{D}\widetilde{Y}],$$

where we assume D cannot be perfectly predicted by W, i.e. $\mathbb{E}[\widetilde{D}^2] > 0$, so β_1 is uniquely defined.

- When we work with a sample, we simply mimic the partialling-out operation in the population. By the FWL Theorem applied to the sample instead of population, the sample linear regression of Y on D and W gives us the estimator

$$\hat{\beta}_1 = \arg\min_b \mathbb{E}_n[(\check{Y}_i - b\check{D}_i)^2] = (\mathbb{E}_n[\check{D}_i^2])^{-1}\mathbb{E}_n[\check{D}_i\check{Y}_i]$$

  which corresponds to the standard ols estimator of β_1.

- **Outlook:** When p/n is not small, using sample linear regression for partialling-out won't be such a good idea and an alternative is to use penalized regression or dimension reduction.

Properties of the estimator

Theorem 3 (Adaptive Inference) Under regularity conditions and if $p/n \approx 0$, the estimation error in $\check{D}_i$ and $\check{Y}_i$ has no first order effect on the stochastic behavior of $\hat{\beta}_1$, namely

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) \approx \sqrt{n}\mathbb{E}_n[\widetilde{D}\varepsilon]/\mathbb{E}_n[\widetilde{D}^2]$$

and consequently,

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) \xrightarrow{\mathrm{D}} N(0, V)$$

where

$$V = (\mathbb{E}[\widetilde{D}^2])^{-1}\mathbb{E}[(\widetilde{D}^2\varepsilon^2)](\mathbb{E}[\widetilde{D}^2])^{-1}.$$

Statistical Inference

- For example, the 95% confidence interval is given by

$$[l, u] := [\hat{\beta}_1 - 1.96\sqrt{\hat{V}/n}, \hat{\beta}_1 + 1.96\sqrt{\hat{V}/n}].$$

  where $\hat{V} \approx V$ is an estimator for the unknown variance of the distribution.

- This set covers the population coefficient β_1 for most realizations of the sample. More precisely, a $(1-\alpha)\%$ confidence interval contains β_1 in approximately $(1-\alpha)\%$ of the realizations of the sample:

$$\Pr(\beta_1 \in [l, u]) \approx 1 - \alpha.$$

Application: Wage Equation

- In labor economics an important question is what determines the wage of workers. Interest in this question goes back to the work of Jacob Mincer (see Lemieux (2006)).
- While determining the inputs that lead to a worker's wage is a causal question, we can begin to investigate it from a predictive perspective. We aim to answer two main questions:

1. The Prediction Question: How can we use job-relevant characteristics, such as education and experience, to best predict wages?
2. The Predictive Effect or Association Question: What is the difference in predicted wages between men and women with the same job-relevant characteristics?[1]

Data (CPS 2015)

- We illustrate using data from the 2015 March Supplement of the U.S. Current Population Survey.
- As outcome, Y, we use the log hourly wage, and we let X denote various characteristics of workers.
- We focus on the (sub) sample of single (never married) workers, which is of size $n \approx 5,000$

Empirical Analysis

- We will estimate a linear predictive (regression) model for log hourly wage using job-relevant characteristics

$$Y = \beta' X + \varepsilon, \quad \varepsilon \perp X.$$

- Then we assess the quality of the empirical prediction rule $\hat{\beta}' X$ using out-of-sample prediction performance, and analyze if there is a difference in pay for men and women (*gender wage gap*).
- The gender wage gap may partly reflect discrimination against women in the labor market (we will discuss this question in more detail later).

Models

We employ two different specifications for prediction:

1. In the *Basic Model* X consists of a set of raw regressors (e.g. gender, experience, education indicators, occupation and industry indicators, and regional indicators), for a total of $p = 51$ regressors. Our basic specification is known in labor economics as the Mincer equation, and is derived from a theoretical economic analysis; see, e.g., Lemieux (2006) for a review.
2. In the *Flexible Model*, X consists of all raw regressors from the basic model as well as *technical regressors*, which are transformations of the raw regressors, namely, polynomials in experience exp^2 and exp^3 and additional two-way interactions of polynomials in experience with all other raw regressors. An example of a regressor created through a two-way interaction is experience times the indicator of having a college degree. In total we have $p = 246$ technical regressors.

Results (Prediction)

- The following table shows the measures of performance.

	p	R^2_{sample}	MSE_{sample}	R^2_{test}	MSE_{test}
basic reg	51	0,31	0,22	0,252	0,245
flexible reg	246	0,35	0,21	0,184	0,267

- The flexible regression model performs slightly better than the basic model on the sample used for training.
- It appears that the basic regression model works slightly better than the flexible regression at predicting log wages for new observations.
- That is, we see that the test (out-of-sample) MSE and R^2 for the basic regression model are respectively slightly lower and higher than those of the flexible regression model, indicating slightly superior out-of-sample predictive performance.
- We will replicate this empirical analysis in the tutorial.

Gender Pay Gap

- An important question is whether there is a difference in (predicted) wages between men and women with the same job-relevant characteristics. To answer this question, we estimate the log-linear regression model:

$$Y = \beta_1 D + \beta_2' W + \varepsilon,$$

  where Y is log-wage, D is the indicator of being female (1 if female and 0 otherwise) and the W's are other determinants of wages: W includes education, polynomials in experience, region, and occupation and industry indicators plus all two-way interactions of polynomial in experience with region, occupation, and industry indicators.

- As we have log-transformed wages, we are analyzing the relative difference in pay for men and women. [1]

Results (Inference)

	Estimate	Std. Error
reg without covariates	-0.038	0.0159
reg with covariates	-0.070	0.0156
partialling-out with covariates	-0.070	0.0156

- We see that the unconditional wage gap of size 3.8% for women increases to about 7% after controlling for worker characteristics. This means we would predict a woman's wage to be about 7% less per hour on average than the wage of a man who had the same experience, education, geographical region and occupation.
- We will have a closer look at this empirical analysis in the tutorial. You can find the notebook here.

Bibliography

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